WORKED SOLUTION

STEP II 1992, Question 6

Pure • Tangents and Root Counting

STEP II 1992, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q6

The key idea

The solutions of $kx = \ln(2\sec x)$ are the points where the straight line $y = kx$ through the origin meets the curve $y = \ln(2\sec x)$. That curve starts above the origin, is convex and shoots off to infinity, so a line through the origin misses it, touches it, or cuts it twice. The special value k_0 is the gradient of the *tangent from the origin*. Tangency means the two curves have the same height and the same gradient, and those two equations give everything else.

Solution

The two graphs

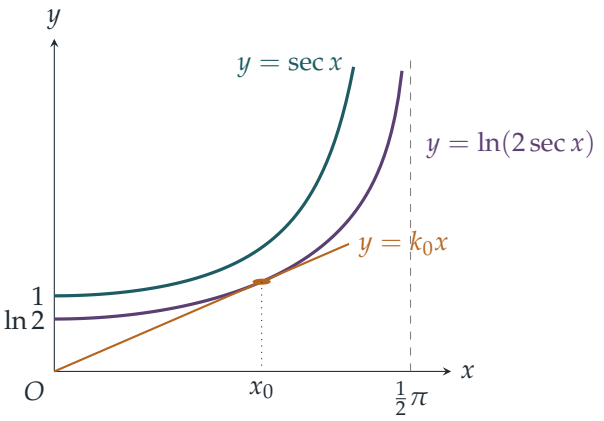
Write $f(x) = \ln(2\sec x) = \ln 2 - \ln\cos x$. Then

$$f'(x) = \frac{\sin x}{\cos x} = \tan x, \qquad f''(x) = \sec^2 x > 0.$$

So $f(0) = \ln 2 \approx 0.69$, $f'(0) = 0$, f is increasing on $(0, \frac{1}{2}\pi)$, it is convex (concave up), and $f(x) \to \infty$ as $x \to \frac{1}{2}\pi^-$ because $\cos x \to 0^+$.

Similarly $y = \sec x$ has $\sec 0 = 1$, gradient $\sec x \tan x$, which is 0 at $x = 0$ and positive afterwards, and $\sec x \to \infty$ as $x \to \frac{1}{2}\pi^-$. Both curves have the vertical asymptote $x = \frac{1}{2}\pi$, and neither is defined there.

The log curve always lies below the secant curve. Putting $u = \sec x \geq 1$, we have $\ln(2u) < u$, because $\ln(2u) - u$ has derivative $\frac{1}{u} - 1 \leq 0$ and so is at most its value at $u = 1$, which is $\ln 2 - 1 < 0$. The log curve also grows far more slowly: as $x \to \frac{1}{2}\pi$ it behaves like ln of the secant.



No solutions for small k, two for large k

Since f is increasing, $f(x) \geq \ln 2$ on $[0, \frac{1}{2}\pi)$, while $kx < \frac{1}{2}\pi k$ there. If $k \leq 2\ln 2/\pi \approx 0.44$, then $kx < \ln 2 \leq f(x)$ throughout, so the line passes underneath the curve and there is no solution.

For large k the line starts below the curve (at $x = 0$ it has height $0 < \ln 2$), but it is steep enough to get above the curve somewhere: at $x = \frac{1}{4}\pi$ the curve has height $\ln(2\sqrt{2}) = \frac{3}{2}\ln 2 \approx 1.04$, and the line is higher there as soon as $k > \frac{6}{\pi}\ln 2 \approx 1.32$. Near $\frac{1}{2}\pi$ the curve escapes to infinity while the line stays at most $\frac{1}{2}\pi k$, so the curve ends up above the line again. The line crosses the curve going up and then again coming down: two solutions.

There cannot be more than two. The function $f(x) - kx$ has second derivative $\sec^2 x > 0$, so it is convex, and a convex function has at most two zeros.

Why k_0 exists

Every solution has $x > 0$, since $f(0) = \ln 2 \neq 0$, so we can divide by x and rewrite the equation as

$$k = g(x), \qquad \text{where } g(x) = \frac{\ln(2\sec x)}{x}, \quad 0 < x < \tfrac{1}{2}\pi.$$

So the number of solutions is the number of times the graph of g reaches height k. Now

$$g'(x) = \frac{xf'(x) - f(x)}{x^2} = \frac{h(x)}{x^2}, \qquad h(x) = x\tan x - \ln(2\sec x),$$

and $h'(x) = \tan x + x\sec^2 x - \tan x = x\sec^2 x > 0$. Since $h(0) = -\ln 2 < 0$ and $h(x) \to \infty$ as $x \to \frac{1}{2}\pi$, h has exactly one zero, x_0. So g decreases on $(0, x_0)$ and increases on $(x_0, \frac{1}{2}\pi)$, and $g \to \infty$ at both ends (the top tends to $\ln 2$ as $x \to 0$, and the top tends to ∞ as $x \to \frac{1}{2}\pi$). Hence, with $k_0 = g(x_0)$, the minimum value of g:

- $k < k_0$: no solutions;

- $k = k_0$: exactly one solution, $x = x_0$;
- $k > k_0$: two solutions.

Graphically, $y = k_0 x$ is the tangent to $y = \ln(2\sec x)$ that passes through the origin. This confirms the sketch argument and pins down the threshold exactly.

Ask yourself Why does "exactly one solution" mean that the line is a tangent?

If the line cut the curve transversally at x_0, then rotating it slightly about O (making k a little smaller or larger) would still give a crossing nearby, and the convexity argument above would give a second one. A single, isolated meeting point that turns into no meeting point when k drops is exactly a point of tangency. Algebraically, this is $h(x_0) = 0$, i.e. $g'(x_0) = 0$.

The equation for x_0

At the point of tangency the line and the curve have the same height and the same gradient:

$$k_0 x_0 = \ln(2\sec x_0), \qquad k_0 = f'(x_0) = \tan x_0.$$

(The second equation is also $h(x_0) = 0$ from above.) Substituting the second into the first gives $x_0 \tan x_0 = \ln(2\sec x_0)$, and since $0 < x_0 < \frac{1}{2}\pi$, $\tan x_0 \neq 0$, so

$$x_0 = \cot x_0 \, \ln(2\sec x_0),$$

as required.

Finding x_0 and k_0

It is tempting to iterate $x_{n+1} = \cot x_n \ln(2\sec x_n)$ directly. That works only slowly: the derivative of the right-hand side at x_0 is $1 - 2x_0/\sin 2x_0 \approx -0.89$, so the iterates oscillate and close in by only about 11% per step. Newton's method on $h(x) = x\tan x - \ln(2\sec x)$ is much better, and $h'(x) = x\sec^2 x$ is very simple:

$$x_{n+1} = x_n - \frac{x_n \tan x_n - \ln(2\sec x_n)}{x_n \sec^2 x_n}.$$

Starting from $x_1 = 1$ (in radians throughout):

$$x_2 \approx 0.9274, \qquad x_3 \approx 0.9147, \qquad x_4 \approx 0.91440.$$

To confirm two decimal places we check a sign change: $h(0.905) \approx -0.023 < 0$ and $h(0.915) \approx +0.0015 > 0$, so $0.905 < x_0 < 0.915$ and

$$x_0 \approx 0.91.$$

Then $k_0 = \tan x_0 \approx \tan 0.9144 \approx 1.30$. Equivalently, $k_0 = \ln(2\sec x_0)/x_0$, which gives the same value.

Checks

- Both expressions for k_0 agree: $\tan(0.91440) \approx 1.2981$ and $\ln(2\sec 0.91440)/0.91440 \approx 1.2981$.
- A numerical scan of $g(x) = \ln(2\sec x)/x$ over $(0, \frac{1}{2}\pi)$ has minimum 1.2981, at $x \approx 0.914$.
- Counting sign changes of $kx - \ln(2\sec x)$: no roots for $k = 1.29$ and two roots for $k = 1.31$.
- The answer fits the rough bounds found earlier: $0.44 < k_0 < 1.32$.

Common slips

- Using degrees. Every value of x here, including the Newton iterates and $\tan x_0$, must be in radians.
- Only one equation for tangency. Meeting the curve is not enough: the gradients must match too, and it is this second condition that pins down k_0.
- Claiming two decimal places without a check. Show a sign change of h across $[0.905, 0.915]$, or iterate until the second decimal place is stable.

Beyond STEP

Counting the roots of $kx = f(x)$ by looking at the range of $f(x)/x$ is a small piece of a big idea. The tangent lines through the origin, and the convexity of f, are what the *Legendre transform* and convex duality are built on, in analysis and in mechanics. Newton's method converges so quickly here because it is quadratically convergent near a simple root, while the fixed-point iteration converges only linearly, with ratio $|\varphi'(x_0)|$. Both facts are proved in a first course in analysis or numerical methods. Bartle and Sherbert's *Introduction to Real Analysis* covers convex functions and the error analysis of Newton's method.

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